CSRD Nature Transition Plan Scorer

Automated ESRS E2–E5 Disclosure Extraction & NTP Scoring

Hackathon CSRD Team

Powered by Azure OpenAI · WWF NAT40 Framework

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What does the system do?

Input

An annual report (CSRD / Universal Registration Document) in Markdown format.

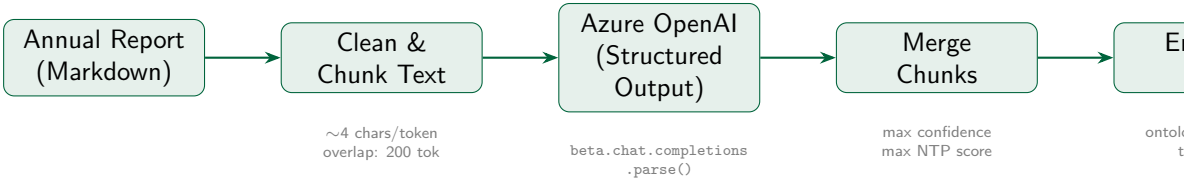
Output (JSON)

- 22 structured ESRS E2–E5 disclosure extractions with evidence & metrics
- 5-dimension Nature Transition Plan (NTP) score modelled on the **WWF NAT40 framework**
- Weighted total score **out of 100**

Why it matters

CSRD mandates disclosure of nature-related impacts (ESRS E2–E5). The WWF NAT40 study shows that only 1 in 40 French large-caps currently meets the average on Nature Transition Plans. This tool makes scoring fast, repeatable, and auditable.

Pipeline Architecture



Chunking strategy
Documents >100k tokens are split into overlapping chunks of 100k tokens with a 200-token overlap to avoid cutting mid-sentence.

Single LLM call per chunk
One `parse()` call returns *both* ESRS disclosures and NTP scores, keeping latency and cost minimal.

Merge strategy
Disclosures: OR on `is_present`, MAX on confidence, union of evidence/metrics. NTP scores: MAX per dimension across chunks.

Technical Stack

Runtime

- Python 3.12
- `openai` SDK — `AzureOpenAI` client
- `pydantic` v2 — schema validation & structured output
- `python-dotenv` — environment configuration

LLM

- Model: `gpt-5-mini` (Azure deployment)
- Endpoint: `acm-openai-tdasc-prd.openai.azure.co`
- API version: `2024-12-01-preview`

Key configuration (`.env`)

```
AZURE_OPENAI_API_KEY=<key>
AZURE_OPENAI_ENDPOINT=https://...
AZURE_OPENAI_API_VERSION=2024-12-01-preview
MODEL=gpt-5-mini
DATA_FOLDER=data
# Optional overrides:
MAX_TOTAL_TOKENS=100000
CHUNK_TOKENS=100000
MAX_GENERATION_TOKENS=20000
TIMEOUT_MS=600000
```

ESRS Disclosure Extraction — Scope

The system extracts **22 disclosure codes** across four ESRS nature standards:

E2 — Pollution

E2-1 Policies
E2-2 Actions
E2-3 Targets
E2-4 Air/Water/Soil
E2-5 Substances
E2-6 Financial effects

E3 — Water

E3-1 Policies
E3-2 Actions
E3-3 Targets
E3-4 Withdrawal
E3-5 Financial effects

E4 — Biodiversity

E4-1 Transition plan
E4-2 Policies
E4-3 Actions
E4-4 Targets
E4-5 Metrics
E4-6 Financial effects

E5 — Resources

E5-1 Policies
E5-2 Actions
E5-3 Targets
E5-4 Inflows
E5-5 Waste
E5-6 Financial effects

Per-disclosure output fields

`code` · `is_present` (bool) · `confidence` [0–1] · `evidence[]{quote}` (verbatim, $\leq$280 chars) · `metrics[]{name, value, unit, year}`

Nature Transition Plan (NTP) Scoring

Based on the **WWF NAT40 methodology** (Feb 2026) applied to CAC 40 companies. Five components scored on a **4-level maturity scale**:

Level	Label	Meaning
0	Non-aligned	Absent or problematic disclosure
1	Compliant	Meets minimum CSRD/ESRS transparency requirements
2	Coherent	Exceeds minimum; integrates WWF/EFRAG expectations
3	Credible	Advanced maturity; science-based frameworks (SBTN, TNFD)

Weighted total score /100:

Dimension	Weight	Max pts
Foundations	25%	25

Formula

$$\text{Score} = \sum \frac{s_d}{3} \times w_d$$

NTP Dimensions — Scoring Criteria

Dimension	Score 1–2 (Compliant/Coherent)	Score 3 (Credible)
Foundations	Double materiality covers direct ops; TNFD or SRTN methodology referenced	Multi-scale analysis (local/regional/global); IPBES pressure categories; KBAs mapped; aligned with Kunming-Montreal
Metrics & Targets	ESRS E2–E5 indicators for direct ops; targets with time horizons	Science-based targets (SBTN-aligned) or contextual/territorial targets linked to state-of-nature indicators
Implementation Strategy	Action plans exist; linked to material impacts; some financial commitments	Site/supply-chain-specific plans; mitigation hierarchy (AR3T); formal financing plan (CAPEX/OPEX)
Engagement Strategy	Named stakeholder groups; described mechanisms	Indigenous & local community guarantees; lobbying positions

Output Format

The pipeline writes `data/output.json` with the following structure:

```
{
  "rick_output_Airbus_2024.json": {
    "disclosures": [
      {
        "code": "E4-2",  "title": "Biodiversity policies",  "standard": "E4",
        "is_present": true,  "confidence": 0.82,
        "evidence": [{"quote": "Our no-net-loss policy applies to all owned sites..."}],
        "metrics": [{"name": "Sites in KBAs", "value": 3, "unit": "count", "year": 2024}]
      },
      ...  // 22 codes total
    ],
    "ntp_scoring": {
      "foundations":              {"score": 2, "maturity": "Coherent", "weight": 25,
                                   "weighted_contribution": 16.7, "rationale": "...", "evidence": [...]},
      "metrics_and_targets":     {"score": 1, "maturity": "Compliant", "weight": 20, ...},
      "implementation_strategy": {"score": 2, "maturity": "Coherent",  "weight": 30, ...},
      "engagement_strategy":     {"score": 1, "maturity": "Compliant", "weight": 15, ...},
      "governance":              {"score": 2, "maturity": "Coherent",  "weight": 10, ...},
      "total_score": 55.0
    }
  }
}
```

Pydantic Data Models

Disclosure models

```python
class Evidence(BaseModel):
    quote: str

class MetricValue(BaseModel):
    name: str
    value: float
    unit: str
    year: Optional[int]

class DisclosureExtraction(BaseModel):
    code: ESRSCode          # E2-1 .. E5-6
    is_present: bool
    confidence: float       # [0,1]
    evidence: List[Evidence]
    metrics: List[MetricValue]
```

NTP scoring models

```python
class NTPDimensionScore(BaseModel):
    score: int    # 0-3
    rationale: str
    evidence: List[Evidence]

class NTPScoringResult(BaseModel):
    foundations: NTPDimensionScore
    metrics_and_targets: NTPDimensionScore
    implementation_strategy: NTPDimensionScore
    engagement_strategy: NTPDimensionScore
    governance: NTPDimensionScore

class CombinedExtractionResult(BaseModel):
    disclosures: List[DisclosureExtraction]
    ntp_scoring: NTPScoringResult
```

`CombinedExtractionResult` is passed directly as `response_format` to `client.beta.chat.completions.parse()`, enforcing strict JSON output.

Project Source Layout

Key files

`main.py`	Entry point
`src/config.py`	`load_config()`
`src/pipeline.py`	`run_pipeline()`
`src/models.py`	Pydantic schemas
`src/prompts.py`	System prompt
`src/esrs_ontology.py`	Code lookup + enrichme
`src/utils/text.py`	Chunking / tokenisation
`src/utils/merge.py`	Multi-chunk merging
`data/input.md`	Input annual report
`data/output.json`	Output

Running the pipeline

```
# Install dependencies
pip install -r requirements.txt

# Run (reads .env automatically)
python main.py

# Override model or chunk size:
MODEL=gpt-4o \
CHUNK_TOKENS=80000 \
python main.py
```

Tip

Set `DATA_FOLDER` in `.env` to point to any directory containing an `input.md` file.

Summary

- ✓ Reads any CSRD annual report in Markdown
- ✓ Chunks large documents with overlap to stay within context
- ✓ Single LLM call per chunk extracts *both* ESRS disclosures and NTP scores
- ✓ 22 ESRS E2–E5 disclosure codes with evidence & metrics
- ✓ 5-dimension NTP score (0–3) aligned with **WWF NAT40**
- ✓ Weighted total out of 100 with maturity labels
- ✓ Fully configurable via environment variables

Reference

WWF France (2026).
The NAT40 Index: CAC 40 Companies Put to the Test by Nature.

ESRS E2–E5, EFRAG (2023).

SBTN, TNFD frameworks.